Quantum Mechanics II — Exercise Sheet 4

Fourier observables on the line

Variant: Solutions

Exercise 1 (An exponential packet). On the line let $\psi(x) = a^{-1/2}e^{-|x|/a}$, where $a > 0$. Using the unitary $\hbar$-Fourier convention, find $\widetilde{\psi}(p)$, ΔQ and ΔP. Decide whether this packet saturates the uncertainty bound. You may use $\int_0^\infty e^{-bx}\cos(cx)\,\mathrm{d}x = b/(b^2+c^2)$ for $b > 0$.

Solution. The position density is even and has integral one. Splitting the Fourier integral into the two half-lines gives

$$\widetilde{\psi}(p) = \sqrt{\frac{2a}{\pi\hbar}}\,\frac{1}{1+a^2p^2/\hbar^2}.$$

The position and momentum means vanish by parity. The second moments are

$$(\Delta Q)^2 = \frac{a^2}{2}, \qquad (\Delta P)^2 = \frac{2\hbar^2}{\pi a^2}\int_{\mathbb{R}} \frac{u^2}{(1+u^2)^2}\,\mathrm{d}u = \frac{\hbar^2}{a^2}.$$

Here $u = ap/\hbar$, and the remaining integral equals $\pi/2$. This computes the variance directly from the Fourier spectral distribution. Thus $\Delta Q\Delta P = \hbar/\sqrt{2} > \hbar/2$, so this packet does not saturate the bound.

Exercise 2 (A bounded momentum window). Let $\widetilde{\psi}(p) = (2b)^{-1/2}\mathbf{1}_{[-b,b]}(p)$ with $b > 0$. (i) Find $\psi(x)$ by Fourier inversion, with its continuous value at zero. (ii) Find its mean free energy and decide whether its position second moment is finite.

Solution. Integrating the exponential over the momentum window gives

$$\psi(x) = \sqrt{\frac{\hbar}{\pi b}}\,\frac{\sin(bx/\hbar)}{x}, \qquad \psi(0) = \sqrt{\frac{b}{\pi\hbar}}.$$

The momentum density is uniform, so

$$\langle H_0\rangle = \frac{1}{2b}\int_{-b}^{b}\frac{p^2}{2m}\,\mathrm{d}p = \frac{b^2}{6m}.$$

For the position second moment the factor x^2 cancels the denominator of $|\psi|^2$. The remaining integral of $\sin^2(bx/\hbar)$ over the line diverges. Thus $\psi \notin D(Q)$, even though every momentum moment is finite.

Exercise 3 (A momentum shift without changing position). For a normalized $f \in \mathcal{S}(\mathbb{R})$ define $\psi(x) = e^{\mathrm{i}p_0x/\hbar}f(x)$, where $p_0 \in \mathbb{R}$. (i) Find its Fourier transform and relate its position and momentum means and variances to those of f. (ii) Explain why a position density alone cannot determine a momentum distribution.

Solution. Combining the phase with the Fourier exponential gives

$$\widetilde{\psi}(p) = \widetilde{f}(p-p_0), \qquad |\psi(x)|^2 = |f(x)|^2.$$

Changing the momentum variable in the moment integrals gives

$$\langle Q\rangle_\psi = \langle Q\rangle_f, \quad \Delta Q_\psi = \Delta Q_f, \quad \langle P\rangle_\psi = \langle P\rangle_f + p_0, \quad \Delta P_\psi = \Delta P_f.$$

Varying p_0 gives different momentum distributions with exactly the same position density.

Exercise 4 (An eigenvalue is not a narrow interval). For the packet in Exercise 2, compute the probability that $|P| \le \varepsilon$ for every $\varepsilon > 0$. (i) Find its limit as $\varepsilon \downarrow 0$. (ii) Compare this with the probability of $P = 0$ and explain why a large probability in a small interval does not produce an eigenvector of P.

Solution. The probability is the length of the intersection of two intervals, divided by $2b$:

$$\mathrm{Prob}(|P| \le \varepsilon) = \min\{\varepsilon/b, 1\}.$$

For this fixed packet it tends to zero as the interval shrinks. The probability of the single value zero is zero because the momentum law has a density. Choosing a different packet with smaller b can concentrate probability near zero, but does not give any one packet an atom there. The momentum multiplication operator has no eigenvector at zero.